

# Spotly — Data Model

## UML Class Diagram

**Course:** ISS296 — Spring 2026  
**Instructor:** Mr. Amine Ben Hassouna  
**Team:** Cyrine Krichen, Eya Sassi,  
Lina Khemiri, Youssef Ben Jemaa

---

## Overview

---

The Spotly data model consists of **nine entity classes** persisted in the browser's local storage. Each class has a dedicated storage key and follows a reactive CRUD pattern. There is no backend: the application is entirely frontend-driven and data is shared across browser tabs via the Web Storage **storage** event.

The diagram uses standard UML class notation. Relationships show the foreign-key field used for association (e.g. **venueId**, **userId**). Composition arrows (*filled diamond*) indicate that child objects have no independent existence outside their parent.

## Persistence keys:

| Class          | Storage Key                    | Notes                                                                                  |
|----------------|--------------------------------|----------------------------------------------------------------------------------------|
| User           | <b>spotly_users</b>            | Email used as natural key                                                              |
| Venue          | <b>spotly_venues</b>           | Linked to User via <b>adminEmail</b>                                                   |
| Environment    | <b>spotly_environments</b>     | Contains embedded <b>FloorElement</b> array                                            |
| MenuItem       | <b>spotly_menu_items</b>       | Scoped to venue; optionally to specific environments                                   |
| Reservation    | <b>spotly_reservations</b>     | Status: REQUESTED   APPROVED   REJECTED   CANCELLED   CHECKED_IN   COMPLETED   NO_SHOW |
| ReservationLog | <b>spotly_reservation_logs</b> | Immutable audit trail; <b>actorRole</b> : client   admin   staff                       |
| Review         | <b>spotly_reviews</b>          | Status: VISIBLE   HIDDEN                                                               |
| VenueStaff     | <b>spotly_venue_staff</b>      | Junction table linking User to Venue as staff                                          |
| WaiterCall     | <b>spotly_waiter_calls</b>     | Status: pending   acknowledged                                                         |

# Class Diagram

